Rami Dmaur

Software Engineer Applied ML

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Poland | Remote | Europe/Warsaw | Full-time or contract | Available ASAP

Profile

Software Engineer with applied ML experience in computer vision, OCR, dataset preparation, model evaluation and mobile inference integration. Developed reinforcement learning models with TensorFlow/Keras and educational vision and sensor classifiers. Combines ML research with application engineering.

Core technologies

Computer vision / OCR, TensorFlow / Keras, TFLite / ONNX Runtime, OpenCV, Dataset evaluation, DQN / PPO, Python

Selected projects

ScanChamp

Software Engineer · Flutter / ML / Backend | ScanChamp | Released | Ongoing development

Computer vision, OCR and on-device inference for a document scanner started from scratch and taken to release; development continues.

- Researched and tuned document-boundary detection, OCR and image-enhancement models; continued this work after the team expanded.
- Created dataset preparation and audit tools tracking document identity, capture session, split and annotation revision; checked for overlap between evaluation splits.
- Compared model and exported-runtime candidates using saved inputs, reports, intermediate images and stage timings; combined metric review with visual inspection of document crops.
- Integrated inference with mobile processing constraints and instrumented model loading, preprocessing, inference, postprocessing and worker round trips.

Technologies: Python, OpenCV, TFLite, ONNX Runtime, OCR, Dataset evaluation, Dart FFI, Mobile inference

Evaluation and runtime tradeoffs

Reference corners are kept separate from detector inputs so annotations do not reveal the answer. Research candidates are compared with release configurations; quality includes preserving document text, while timing is broken down by processing stage. Experimental models and diagnostic resources remain separate from store defaults.

Octy Marketplace

Flutter Team Lead · Backend ownership | Octy Marketplace | Pre-production | Funding ended

Reinforcement-learning research for a marketplace loyalty and discount system, evaluated in simulation before production.

- Developed a DQN approach in TensorFlow/Keras with replay buffer and target network, together with a custom Gym environment.
- Implemented a separate PPO actor/critic approach using GAE and a clipped objective; compared policies through simulations and analysis.
- Explored reward formulation for loyalty/discount decisions and built research tooling for inspecting training and policy behavior.

Technologies: Python, TensorFlow, Keras, DQN, PPO, Gym, NumPy, Pandas, Streamlit, Matplotlib

Research context

The experiments were connected to the marketplace product and its loyalty scenarios. The project stopped before production; policy behavior was evaluated in a simulation rather than through live customer discounts or commercial metrics.

Foot Neuro

Python / ML Developer | Academic project

Academic image-classification prototype for foot-condition assessment with batch processing and per-image confidence.

- Compared MobileNetV2, ResNet50 and EfficientNetB0 using transfer learning, a frozen base and subsequent fine-tuning.
- Used tf.data augmentation, Dropout, L2, early stopping and learning-rate scheduling; explored hyperparameters and cross-validation.
- Worked with KFold/StratifiedKFold and confusion matrices; integrated TensorFlow/Keras predictions and confidence into a PyQt6 batch interface.

Technologies: Python, TensorFlow, Keras, Transfer learning, scikit-learn, Cross-validation, tf.data, NumPy, PyQt6

Tremor screening

Flutter / Backend / ML Developer | Academic project

Academic binary-classification prototype using smartphone accelerometer and gyroscope data for a screening recommendation.

- Built sensor collection in Flutter and a Flask API for preparing the recorded data.
- Developed a TensorFlow/Keras classifier and returned its screening recommendation to the application.

Technologies: Python, TensorFlow, Keras, Sensor data, Flask

CT Teacher

Full Stack Developer | Personal project

Personal research into AI-generated exercises, video lessons, speech and educational games.

- Explored exercise and analogous-task generation and integrated generated content with the learning platform.
- Assembled video-lesson and voice-generation workflows with rendering tools and ffmpeg; investigated speech models and ONNX/MLX-based tooling.
- Worked through content checking, formula rendering, pronunciation and video assembly within an educational research prototype.

Technologies: Python, AI / LLM integration, ONNX Runtime, MLX, Hugging Face tooling, ffmpeg, FastAPI

Employment history

ScanChamp | Mar 2026 — present
Software Engineer · Flutter / ML / Backend

Project-based development | Aug 2025 — Mar 2026
Senior Flutter Developer

Octy Marketplace | Dec 2024 — Aug 2025
Flutter Team Lead · Backend ownership

Success Craft | 2024
Senior Flutter Developer

Developers Heaven | Within 2022–2024
Flutter Developer

Contract engagements | May 2022 — Dec 2024
Senior Flutter Developer / Team Lead

iTechArt Group | May 2021 — May 2022
Junior–Middle Flutter Developer

Gieksby Development / Secreate | Oct 2020 — May 2021
Junior Flutter / React Native Developer

SmartAppTechnology Limited | Jul 2019 — Oct 2020
Full Stack Web Developer

Education and languages

BSUIR, 2023 · Faculty of Computer-Aided Design · Electronics and Software Engineering

Russian — native; English — B2; Arabic — A1